Tokenjet Roadmap (18 Months)

A study-and-build plan for Tokenjet, an LLM inference engine with continuous batching, a paged KV cache, quantization and speculative decoding.

Rule for every phase: study for the first third, build for the rest, and end with a written note in `docs/`.

Phase 1: Foundations (months 1-3)

Study

- Python and PyTorch fundamentals (the official PyTorch tutorials).
- Andrej Karpathy's "Neural Networks: Zero to Hero", especially "Let's build GPT".
- "Attention Is All You Need" and Jay Alammar's *Illustrated Transformer*.
- Basic GPU concepts: memory, bandwidth, and why generation is memory-bound.

Build

- Write a tiny GPT from scratch and generate text with it.
- Load Gemma (2B) with Hugging Face and measure baseline speed and memory.

Deliverable: a baseline benchmark script and a note on where time and memory go.

Phase 2: Core engine (months 4-6)

Study

- How autoregressive decoding works: the prefill and decode phases.
- KV cache: what it stores and why it grows.
- Read the llama.cpp and nanoGPT code.

Build

- Your own generation loop with a KV cache, a simple request queue and a scheduler.
- A FastAPI server with streaming output.
- Unit tests that check your output matches Hugging Face.

Deliverable: Tokenjet v0.1, a working single-model server.

Phase 3: Optimization (months 7-10)

Study

- The Orca paper (continuous batching).
- The vLLM paper (PagedAttention).
- LLM.int8, GPTQ and AWQ (quantization).

Build

- Continuous batching.
- A paged KV cache with block allocation.
- INT8 or INT4 quantization.

Milestone (month 10): continuous batching and the paged cache work and beat your baseline. Don't move on without this.

Phase 4: Advanced (months 11-13)

Study

- Speculative decoding (the Leviathan et al. and Chen et al. papers).
- FlashAttention.
- Triton tutorials and the GPU MODE lectures.

Build

- Speculative decoding with a small draft model.
- One custom Triton kernel, such as a fused attention or cache operation (stretch).

Phase 5: Benchmarking (months 14-15)

- Compare against Hugging Face and vLLM on the same GPU.
- Measure throughput, latency (time to first token and per token) and memory at several batch sizes and loads.
- Run ablations that switch each optimization on and off.
- Keep a reproducible script for every chart.

Phase 6: Writing and release (months 16-18)

- Write the paper or technical report, your KTU final report, and a clean README with charts.
- Prepare a demo and a 10-minute talk.
- Optionally make an upstream contribution to vLLM or llama.cpp.

Habits throughout

- Keep a weekly log in `docs/log.md`, make small commits, and review each other's pull requests.
- Split the work: one of you leads the engine and kernels, the other leads the server, benchmarks and docs. Swap at each phase boundary.
- Check KTU's final-project deadlines and mark them on this timeline now.